8 Features of manufacturing industries

8.2 Quality Monitoring Systems

Topic	What students need to learn:
8 Features of manufacturing industries	8.2 – Characteristics, application, advantages and disadvantages of the following types of quality monitoring systems: a) ***quality control*** *– the monitoring and achieving of high standards and degree of tolerance by inspection and testing, computer-aided testing* b) ***quality assurance*** *– monitoring the quality of a product from its design and development stage, through its manufacture, to its end-use performance and degree of customer satisfaction*

Topic	What students need to learn:
8 Features of manufacturing industries	8.2 – Characteristics, application, advantages and disadvantages of the following types of quality monitoring systems: *c)* **Total Quality Management (TQM)** – when applied to quality assurance procedures and its impact on employees at every stage of the production process, ISO 9000.

Quality Control (QC)

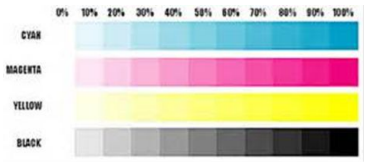
Quality Control (QC)

Quality Control (QC) is the monitoring, checking and testing of QA tolerance conformity throughout production specified by strict guidance documentation of client requirements.

Monitoring, Checking and Testing

- Depending on the nature of the product being manufactured, the level of inspection may be:
  - *Full (100%) inspection of every unit (e.g. for a safety critical component of a jet aircraft)*
  - *Inspection of samples at predetermined intervals (e.g. mass-produced PET drinks bottles)*

- Compliance of materials with the manufacturer's specification can be checked by:
  - *Destructive testing of samples to assess properties such as tensile and compressive strength*
  - *Simple visual inspection*
  - *Chemical analysis of samples (e.g. moisture content or chemical composition)*
  - *Colour matching for pigments or paints*



Monitoring, Checking and Testing

- Dimensional accuracy can be checked by:
  - *Use of digital measuring devices such as micrometers and digital Vernier calipers*
  - *Interval sample testing may be conducted in a QC department.*

Tolerance

- ***Tolerance*** is the level of accuracy acceptable in a particular situation.
- It depends on the material used for parts, as well as their size and function.
- The wide tolerance acceptable for a potter throwing a fruit bowl in clay contrasts with the very fine one that is essential when precision engineering a complex turned part for a car gearbox.
- Fine tolerances are particularly important when interchangeability of parts is required.

Tolerance

- Normally expressed as X ± Y (X = exact size required and Y = allowable error).
- Components not within tolerance are normally rejected by QA and QC measures.
- ***Go/No-Go gauges*** are used to check specific tolerances in mass production.

Specific Quality Control Methods

Specific Quality Control Methods

Go/No-Go Gauge

- These check a single measurement for tolerance range, with a pass/fail reading.
- You should have covered these in *chapter 3.2 – Applications of specialist measuring tools and equipment.*

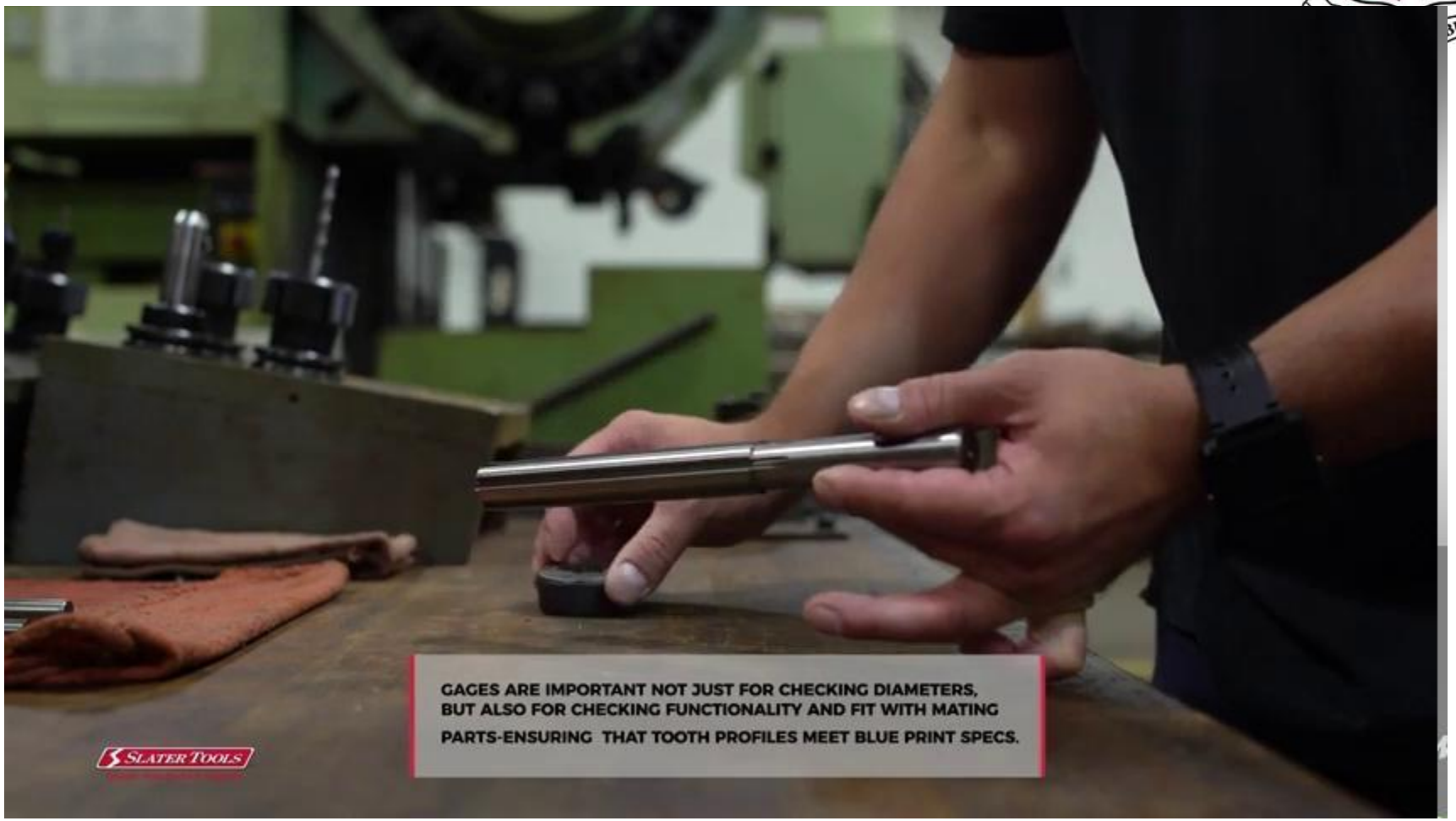








Video: Go/No-Go Gauge

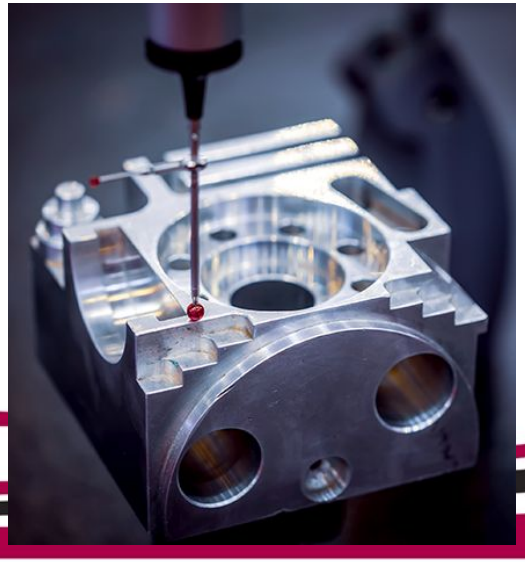


Specific Quality Control Methods

Laser or Probe Scanning and Measuring

- Computer-Aided inspection usually involves **co-ordinated measuring machinery (CMM)**.
- Probe scanners can check predefined measurements of finished components.
- CMM is used to check tooling accuracy for QA tool change procedures.





Video: Co-ordinated Measuring Machinery (CMM)



Specific Quality Control Methods

Laser or Probe Scanning and Measuring

Advantages:

- Highly accurate, with automatic comparison with computer 3D CAD model.
- Non-contact lasers can scan thousands of readings per second for greater accuracy.

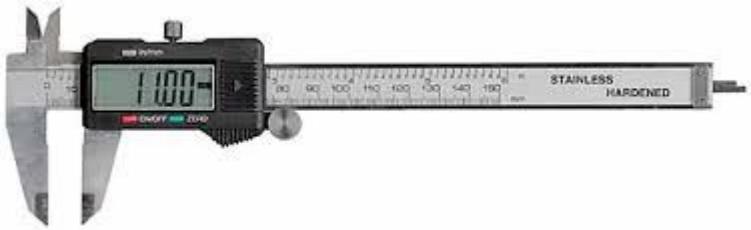
Disadvantages:

- Expensive initial cost for setting up links to IT systems.
- Need for highly trained personnel.
- On-going maintenance and calibration requirements.

Specific Quality Control Methods

Digital Measuring Devices

- Digital Vernier calipers and micrometers provide a readout of the dimension being measured.
- A common application is for ‘interval’ production line sample testing in a quality control department.
- Vernier calipers are generally used for external, internal and depth measurement readings with ±0.02mm accuracy.





Specific Quality Control Methods

Digital Measuring Devices

Advantages:

- These are ‘flexible’ measuring tools as they facilitate dimensional accuracy checks in a range of situations.
- The screw thread utilised in a micrometer facilitates accuracy of ±0.01mm, with some capable of ±0.001mm.
- They are obtainable in a range of sizes and configurations.

Disadvantages:

- Training is needed, so unskilled personnel can’t use them.
- They are precision instruments and need to be carefully maintained and handled.

Quality Assurance (QA)

Quality Assurance (QA)

Quality Assurance (QA) is a system of procedures and policies to ensure products are ***'right first time and every time'***, to eliminate faulty products and ensure customer satisfaction.

Quality Assurance (QA)

Examples of QA procedures and policies:

- Only use materials and components from **ISO 9000 Quality Management Standard Suppliers**.
- Checking correct temperatures for effective cavity filling in moulding procedures.
- Rigid machinery and cutter maintenance schedules to maintain tolerances.
- Effective QA procedures reduce the number of QC issues and rejects.
- Pre-production ***CAD modelling*** to ***check fit*** and ***assembly***.
- **FEA (finite element analysis)** to analyse the effect of stresses.
- **MFA (mould flow analysis)** to predict material flow within moulding machinery.
- Incorporation of embossed ***mould identification marks*** for ***tracing fault***s.

Video: ISO 9000 Quality Management Standard



Total Quality Management (TQM)

Total Quality Management (TQM)

- ***Total Quality Management (TQM)*** aims to remove waste and make products right first time by ***continuous improvement***.
- The workforce's views, feedback, teamwork and communication are highly prioritised.
- A culture of high-quality is the predominant feature of all aspects of production.

Video: Toyota - Total Quality Management (TQM)



Quality Standards

Quality Standards

- ***ISO 9000*** is the most important quality standard to guide manufacturers and other organisations.
- This quality management system guides organisations in satisfying the requirements of customers and other stakeholders who interact with its products or services.
- Companies who follow this globally recognised standard are at an advantage when seeking contracts, and ISO 9000 certification is sometimes a compulsory requirement.
- Conforming to regular updated national, European and international standards contributes to the commercial success of products. We looked at these in chapter 11.4.

Exam Tip:

Questions requiring information about how you organised your NEA to ensure a good-quality outcome shouldn't be just a diary-style list of processes.

It is essential to include reference to the *implementation and organisation of quality control measures*.

Test Yourself:

1. *Explain **<u>two</u> QA procedures** that may be used to **reduce defects** when injection **moulding** components.*
2. *What is a **go/no-go gauge** and where might you use it?*
3. *Name a **digital measuring device**, give an example of where it might be used and explain why it is regarded as a **'flexible'** measuring device.*
4. *Briefly explain the term **'tolerance'** and contrast its importance for making a fireplace poker and a ball for the tip of a biro.*
5. *How might a **destructive** quality laboratory control test be used to ensure that the material being used to make a carrier bag handle will not snap when subjected to a heavy load of shopping?*
6. *What are the benefits for a car component manufacturer of complying with the **ISO 9000** series of quality standards?*